

Development of a Gray-box Modeling Method for HVAC&R Systems in Commercial Buildings with an Automatic Calibration Process

Hyemi Kim¹, Sumiyoshi Daisuke¹, Youngjin Choi²

¹Kyushu University, Fukuoka, Japan

²Kyonggi University, Suwon City, Gyeonggi-do, Republic of Korea

Abstract

A gray-box model overcomes the limitations of black-box building modeling. However, existing gray-box models have many parameters and calculate an optimized objective function from the data. We developed gray-box techniques for heating, ventilation, air-conditioning, and refrigeration (HVAC&R) equipment to understand the status of the system operation. The parameters of the individual equipment model are minimized and fitted for interval data by a machine learning model to automatically perform the calibration process. The power consumption of the individual models was approximately 14% higher than that of TRNSYS, and the power consumption of the entire system model was approximately 7% lower.

Key Innovations

- Modeling is possible using filed data without preprocessing
- Modeling is possible using only the flow rate and temperature data in the system by minimizing the input of equipment information (catalog values, etc.)
- Reduce the number of parameters and minimize the modeler's subjective intervention
- The calculation is possible without calibration by calculating the parameters for interval data (e.g. 1 min.): Live parameterization
- System operation can be simulated using only data without 3D modeling of buildings and systems

Practical Implications

The developed simulation tool can reproduce the system operation status using the flow rate and temperature data. Raw data can be used without normalization. Various characteristics collected from different systems can be analyzed. A close view of the device's performance and system operation is provided.

Nomenclature

T_{cond}^{in} = Condenser Inlet Water Temperature (°C)
 T_{evap}^{in} = Evaporator Inlet Water Temperature (°C)
 Q_{evap} = Cooling load (kW)
 $Q_{leak,eqv}$ = Evaporator Heat Leak (kW)
 ΔS_T = Internal entropy generation (kW/K)
 R = Thermal resistance (K/kW)
 $T_{cond,TRNCata}^{in}$ = Condenser Inlet Water Temperature of TRNSYS Catalog (°C)
 $Q_{evap,TRNCata}$ = Cooling load of TRNSYS Catalog (kW)

$COP_{TRNCata}$ = Coefficient of Performance from TRNSYS Catalog (-)

COP_{TRNSYS} = Coefficient of Performance from TRNSYS Simulation Result (-)

P_R = Power of Heat Source Equipment (kW)

T_a = Outdoor Temperature (°C)

$Power_{pump}$ = Power of Pump (kW)

η = Efficiency of Pump (%)

Q = Discharge Head (m³/s)

ρ = Density of Water (kg/m³)

H = Total Pump Head (m)

Q_H = Heat exchanger

T_{h1} = Hot Side Inlet Temperature (°C)

T_{h2} = Hot Side Outlet Temperature (°C)

T_{c1} = Low Side Inlet Temperature (°C)

T_{c2} = Low Side Outlet Temperature (°C)

K = Heat Transmission Coefficient (kcal/(m² h °C))

A = Surface Area (m²)

$\dot{m}_{evap}$ = Flow rate into the Evaporator (m³/h)

T_{evap}^{out} = Evaporator Outlet Water Temperature (°C)

COP = Calculated Coefficient of Performance (-)

$Power_{chiller}$ = Power of Chiller (kW)

T_{pump}^{in} = Pump Inlet Water Temperature (°C)

$\dot{m}_{pump}$ = Flow rate into the Pump (m³/h)

$\dot{m}_{source}$ = Hot Side Flow Rate (m³/h)

$\dot{m}_{load}$ = Low Side Flow Rate (m³/h)

Introduction

Many data-processing technologies have recently been developed. In addition, the accuracy of the sensor is improved; therefore, offline data can be collected with a small discrepancy. Hence, a large amount of collected data is being used in various industries. For instance, the installation of an energy management system to monitor the actual building status and reduce the energy consumption is increasing in the building energy field (ABB Group 2016).

The black-box method is a data-driven technique that applies inverse modeling based on the collected data, where the causal relationship in the calculation process cannot be fully recognized. This method is currently being actively used in building energy simulations (Abdul, 2014). The calculation used with this method is applied swiftly based on the collected data; hence, a subjective

judgment by the modeler is minimized. However, this method has problems related to the reliability of the collected data and data loss during the collection process. Because the black-box method is not physical, the calculated results may be irrelevant to the actual building operation status. It is also difficult to conduct calculations when the amount of data collected is small or when the newly collected data are outside a particular range (Harish, 2016).

In this study, to avoid the problems of the black-box method, a gray-box method (a hybrid method) using input/output variables and parameters derived from physical equations was developed. Furthermore, using the concept of the existing gray-box method, an inverse modeling simulation tool that uses only the data that reflect the physical characteristics of the system was developed. Therefore, the operating status of the system can be extracted from the collected data to determine whether there is an abnormality in the behavior of the system.

Among the studies that apply the gray box model in the field of building energy simulations, Afram (2015), Coninck (2014), and Sawant (2020) conducted research on residential buildings. All studies used a measured dataset, and the room temperature and energy consumption were calculated using these data. These studies are aimed at finding a way to reduce energy consumption through optimal control. Although the modeling result shows a high accuracy, the number of parameters to be estimated is large for each piece of equipment, and modeling is conducted in combination with existing simulation tools (e.g., Energy Plus and Modelica).

Prior studies on the gray box model of commercial buildings (Pugliese, 2016, Takagi, 2015, Braun, 2002) have mainly focused on a method of directly predicting the secondary load and calculating the energy consumption of the entire building. The modeling of the HVAC&R system was simulated using existing tools. Modeling studies using a data-driven technique for the HVAC&R system of commercial buildings have yet to be studied in detail.

Therefore, studies on a gray-box model for the primary side of HVAC&R systems of commercial buildings have been conducted. Lee (2010), Estrada-Flores (2006), and Hydeman (1995) developed gray-box models for heat source equipment such as chillers and heat pumps. Arahall (2008), Berkenkamp (2014), and Tardioli (2020) developed heat storage tanks, and Singh (2016) developed a cooling tower model. The above studies described the development of a modeling method based on the equipment rather than the system operation. Consequently, to reduce the energy consumption of commercial buildings, studies on a gray-box modeling method that can obtain the detailed performance of each HVAC&R system and the current operation status of the entire system are required.

Furthermore, most studies have calculated optimized parameters used in the modeling process and applied them

as fixed values. In this case, because fixed data are substituted, the error may increase owing to data fluctuations. However, when the parameters are calculated in real time, a prediction result with high accuracy can be calculated without calibration.

In this study, we developed an automatic simulation tool for the HVAC&R system (on the primary side) for commercial buildings using a data-based gray-box modeling method. The aim was to minimize the burden on the modeler based on the data collected using the actual measurement of the HVAC&R system and automate the entire simulation process. First, we developed an inverse modeling technique for HVAC&R systems in which an automatic calibration (or non-calibration) was conducted. This is because the calculation method developed in this study calculates the parameters according to all data intervals, and thus predicts the parameters in real time.

Our research was conducted by creating a simplified system to examine the feasibility of the proposed technique. A simple HVAC&R system was created using TRNSYS, and virtual data on the operation of the cooling period were prepared. Subsequently, the gray-box modeling method was developed and verified for each piece of equipment using the data, and the final verification was performed by expanding it to the system range. To model each piece of equipment, the parameters were derived from the physical equation considering the measurement method and limitations. Parameter training was conducted using an artificial neural network (ANN) model, and individual equipment parameters were predicted for the newly collected data, and the accuracy was verified. In addition, the system operation calculation was conducted by obtaining the input and output relationship data for each piece of equipment.

This method allows us to obtain the detailed operation status of the HVAC&R system through individual equipment modeling, and to comprehend the physical relationship based on the correlation between the equipment through the operation results of the entire system. In addition, the simplicity and calculation speed of the modeling process can be increased by minimizing the required input variables and parameters. Even when using field data, modeling is possible with only the flow rate and temperature of the equipment, and thus the direct intervention of the modeler is significantly reduced. This means that the modeler workload and subjective intervention are minimized. Moreover, as the parameters were predicted for each time depending on the driving conditions, the simulation results could be automatically corrected. In addition, the actual operation status of the target system can be reflected because it is based on the characteristics of the data collected.

Overview of Simulation Tools

Research Method

Figure 1 shows the sequence of the study. First, the dataset of the simplified system was simulated using TRNSYS (Figure 1(a)), and the value was considered as the field data. The next step is a parameter selection step (Figure 1(b)), which calculates a target value (e.g., a parameter)

based on the physics formula by using the data collected. During the training step (Figure 1(c)), each target value of the equipment, which is a sub-model, is fitted using a dataset from 2016, and the ANN model with the highest accuracy of the validation data is saved. Subsequently, to derive a prediction value for the 2017 dataset (Figure 1(d)), an instantaneous calculation of the entire system is possible for each data interval (1 min) using the individual sub-ANN models with the highest accuracy.

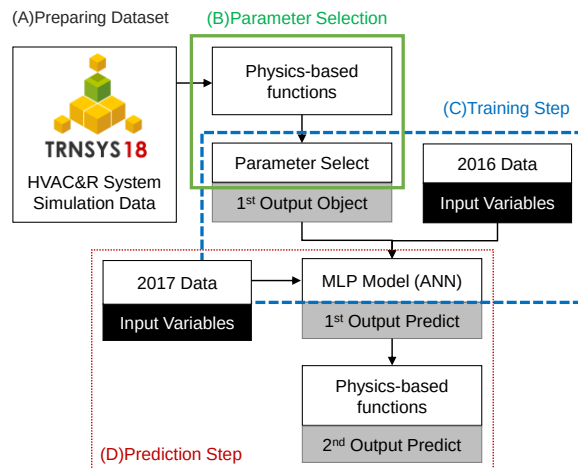


Figure 1: Flow of Research in this paper.

When creating a simple system, the cooling load of the building is required. This study used actual data collected from a building energy management system (BEMS). The BEMS data were collected from an actual building with five floors above the ground and a total floor area of 11,613.3m² over a 3-month period from July to September in 2016 and 2017, measured at 1-min intervals. Accordingly, the simulation value of TRNSYS was also generated for the following three months each year. Finally, to use only the stabilized simulation results for TRNSYS, 12,440 data points (per year) from July 8 to September 30, excluding the first weeks, were trained (2016) and predicted (2017). Each data point was used in the verification step, where 20% (62,200 data points) of the training data were used. As such, this study uses virtual data, and thus it does not consider problems such as sensor errors and data losses that may occur when collecting actual data.

A review of the input variables (Figure 1(c)) was conducted to find the optimized ANN model for each sub-model during the training stage. In the case of adding items such as control signals that can be collected from the system in addition to the temperature and flow rate derived from the physics formulas, the modeling results for each equipment were verified, and the system operation was predicted using the calculated high-performance ANN model. We used the root mean square error (RMSE) and R-squared (R^2) for individual equipment modeling and verification of the results of the system operation.

HVAC&R System Overview

This study adopts a central heat source system that handles cooling loads using air as a heat source and reduces peak loads during the day using the charged heat

of the storage tank. The chilled water loop (Figure 2) had two air-cooled turbo chillers and one heat storage tank, and the second chiller was used for a heat storage operation conducted at night. When a cooling load occurred, the chilled water was first radiated through the heat exchanger, and after the cooling load of the building exceeded 350 kW, one chiller was additionally operated. All pumps had a variable flow, and a bypass pipe was placed to control the flow rate and temperature sent to the secondary side. In this study, a gray-box modeling technique was introduced for each of two air-cooled turbo chillers, a heat exchanger, and five pumps, and a physical storage tank model was added to calculate the overall system calculation reflecting the results of a heat storage tank under a steady state without heat loss. That is, the modeling development targets of this study are limited to air-cooled turbo chillers, heat exchangers, and pumps.

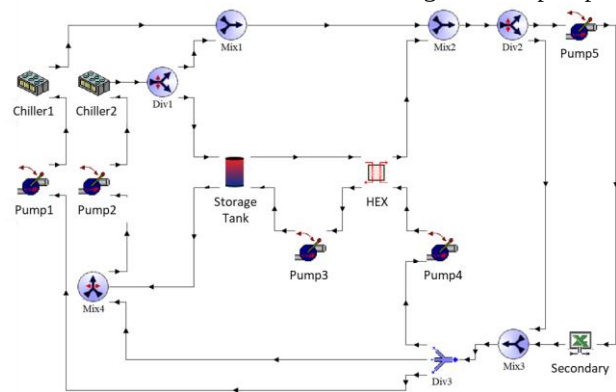


Figure 2: Schematic of TRNSYS HVAC&R System.

System Modeling Method

The calculation process for the entire system is shown in the prediction step in Figure 3. Figure 3 (a) shows the construct of the entire sub-model, and Figure 3 (b) shows the prediction process of the complete tool. This tool has four sub-models: building load, flow rate, equipment, and connection models, each of which uses the ANN method. First, training was conducted for each sub-model, and the optimal model was stored. Subsequently, the parameters were predicted for each data interval, and the outlet temperature and power amount were calculated by substituting them into the theoretical equation. The outlet temperature calculated from one equipment model becomes the input value of the other equipment for the next time calculation. For example, the predicted chilled water outlet temperature of chiller 02 is added to the low-side inlet temperature of the heat storage tank. To date, these inlet and outlet temperature calculations have been passively conducted on manual operations; however, in the future the “connection sub-model” developed will be completed, and the connection between the inlet and outlet temperatures will be automatically determined and calculated. In addition, the section of the mixed temperature is determined based on the results of the passive connection model, and the flow rate and temperature transmitted to the secondary side are calculated. Finally, the outlet temperature and power of each piece of equipment, chilled water supply, and return temperature are verified.

Equipment Modeling

The equipment used in the system include a heat source equipment, heat storage tank, pump, and heat exchanger. The modeling constraints are as follows:

- The number of parameters is minimized.
- When selecting the parameters, we considered the parameters that changed each time interval based on the physical equations and items that were impossible to measure or were not generally measured (e.g., heat loss, pump efficiency, and head).
- The minimum number of input variables should be selected when considering the limitations of the collected data; however, the input variable should be based on physics formulas.
- It was assumed that no heat loss occurred in the hydronic system.

The above selection condition resulted in calculations with the fewest input variables, considering that the possible measured input variables were limited and the measured items were different for each building. The minimum required input variables were the flow rate and temperature for each piece of equipment used in the entire system calculations. In addition, the intention was to enable a real-time simulation in the future using the shape of the collected data as much as possible during the modeling.

Heat Source Equipment Model

A TRNSYS Type 118 Air-Cooled Chiller was adopted as the heat source in this study. First, the chilled water outlet temperature parameter (hereinafter, $Pmt_{T_{evap}^{out}}$) for calculating the chilled water outlet temperature is determined using (1). The equation was studied by

Gordon (1995) and is called the simple thermodynamic model.

$$\frac{T_{evap}^{in}}{T_{cond}^{in}} \left(1 + \frac{1}{COP}\right) - 1 = \frac{T_{evap}^{in}}{Q_{evap}} \Delta S_T + Q_{leak,eqv} \frac{(T_{cond}^{in} - T_{evap}^{in})}{T_{cond}^{in} \times Q_{evap}} + \frac{R \times Q_{evap}}{T_{cond}^{in}} \left(1 + \frac{1}{COP}\right) \quad (1)$$

$$Pmt_{T_{evap}^{out}} = T_{evap}^{in} \cdot P_R - \Delta S_T \cdot T_{evap}^{in} \cdot T_{cond}^{in} - Q_{leak,eqv} (T_{evap}^{in} - T_{cond}^{in}) \quad (2)$$

Furthermore, using the catalog value in TRNSYS, the relationship between the cooling capacities, the inlet temperature on the heat source side, and the coefficient of performance (COP) is generated by (3), which is a multiple regression equation.

$$COP_{TRNCata} = \beta_0 + \beta_1 T_{cond,TRNCata}^{in} + \beta_2 Q_{evap,TRNCata} + \beta_{11} T_{cond,TRNCata}^{in^2} + \beta_{22} Q_{evap,TRNCata}^2 + \beta_{12} T_{cond,TRNCata}^{in} Q_{evap,TRNCata} + \varepsilon \quad (3)$$

$$COP_{TRNSYS} = COP_{TRNCata} \times Pmt_{COP} \quad (4)$$

The discrepancy between the catalog values of the COP generated using (3) and the COP during the actual operation (i.e., COP of TRNSYS result) is set as the COP parameter (hereinafter, Pmt_{COP} , according to (4)). In addition, $Pmt_{T_{evap}^{out}}$ and Pmt_{COP} are trained using the MLP model, and the two parameters are then predicted using the data on the prediction period. By substituting the predicted parameters into the theoretical formula for calculating the performance of the heat source equipment, the cold-water outlet temperature, COP, and power are calculated. The four input variables are the inlet temperature on the heat source side, inlet temperature on the load side, flow rate on the heat source side, and flow rate on the load side, considering the characteristics of the

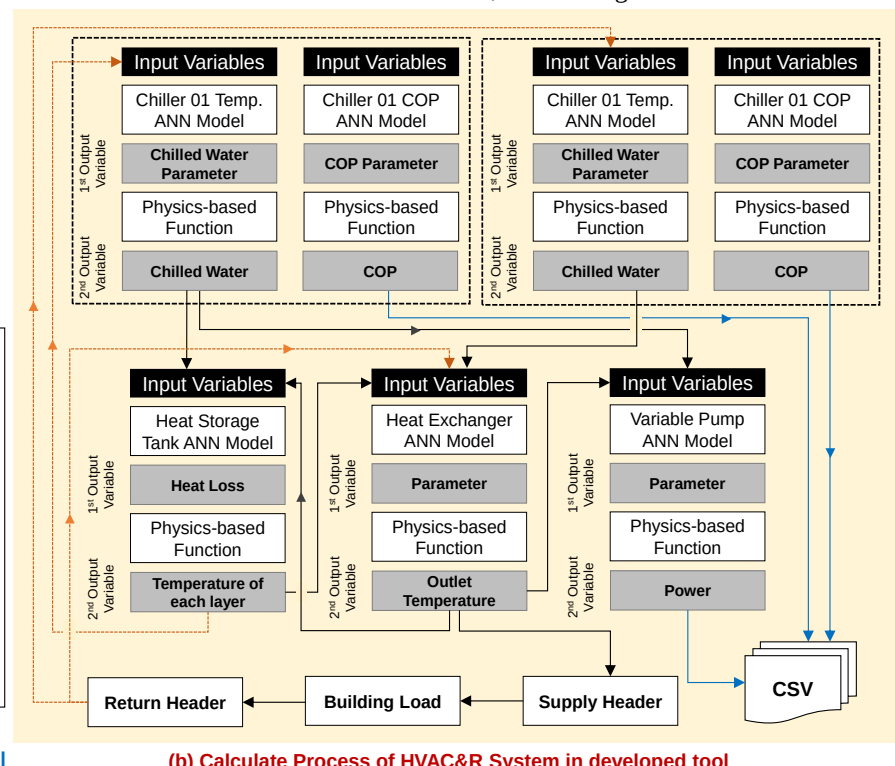
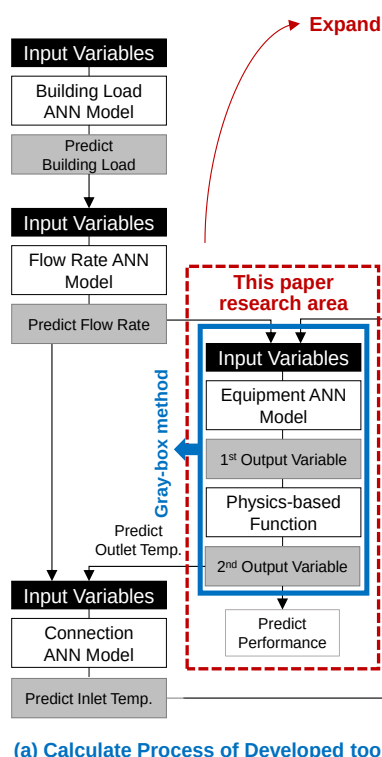


Figure 3: Process of entire simulation method.

theoretical equation and the measured data. Therefore, we set the three input variables.

Pump Model

The types of pumps are largely divided into a constant flow and variable flow pumps. In this study, type 741 was adopted as the variable flow pump. The theory for calculating the shaft power of the pump is given by (5). The efficiency and head of the motor act as parameters in the following equation:

$$Power_{pump} = \frac{\rho Q H}{1.02 \eta} \quad (5)$$

$$Pmt_{pump} = \frac{H}{\eta} \quad (6)$$

The water temperature and flow rate flowing into the pump are input variables based on the physical equation.

Heat Exchanger Model

Parallel and counter flow types exist as heat exchange methods for the heat exchanger. In this study, a Type 5 counter-flow heat exchanger, which is applied as a counter-flow-type heat exchanger, was adopted. The calculation of the counter-flow type is given in (7). In addition, $KA(Pmt_{HEX})$ was selected as a parameter whose value changes over time by the variables input during the heat exchange. The inlet temperature and flow rate of the heat source and load sides were set as the four input variables based on the following theoretical equation:

$$Q_H = KA \frac{(T_{h2} - T_{c1}) - (T_{h1} - T_{c2})}{\log \frac{T_{h2} - T_{c1}}{T_{h1} - T_{c2}}} \quad (7)$$

Table 1: Input and output variables of each equipment.

Equipment	Input	1 st Output	2 nd Output
Heat Source Equipment	Evaporator Inlet Water Temp.	$Pmt_{T_{evap}}$ Pmt_{COP}	Evaporator Outlet Water Temp.
	Condenser Inlet Water Temp.		COP
	Flow rate into the Evaporator		Power
Pump	Inlet Water Temp.	Pmt_{pump}	Power
	Flow Rate		
Heat Exchanger (Water to Water)	Hot Side Flow Rate	Pmt_{HEX}	Hot Side Outlet Temp.
	Low Side Flow Rate		Low Side Outlet Temp.
	Hot Side Inlet Temp.		
	Low Side Inlet Temp.		

The modeling method for each piece of equipment is as described above, and the input and output variables for each equipment are summarized in Table 1. The proposed modeling method takes an 1st output variable as a parameter and applies the fitting using the MLP model. Using the calculated parameters (1st output in Table 1), the performance of each piece of equipment was finally

calculated by substituting into the theoretical equation, which is indicated as the “2nd Output” in Table 1.

The gray-box modeling method of the proposed HVAC&R system was compared to the existing gray-box modeling methods (Lee, 2010; Estrada-Flores, 2006; Hydeman, 1995) as follows: The existing gray-box modeling method was employed by creating an objective function that approximates the correlation in the collected data and determining the coefficients in the objective function. With this method, a plurality of coefficients should be calculated to minimize errors; however, once the coefficient is calculated, the same coefficient is continuously applied. Even in the same system, it is difficult to find the optimal coefficient flexibly when data are continuously added. Using our physical formula calculation method to calculate the minimum value of the parameter, a flexible response to changes in data is possible.

Artificial Neural Network Model

The multilayer perceptron (MLP) model was adopted as a calculation method for predicting the parameters. By stacking hidden layers in multiple layers, nonlinearly separated data can be learned. Table 2 presents the algorithm for the MLP model.

Table 2: Algorithms of MLP Model of Each Equipment.

PSEUDO CODE OF MLP MODEL
Inputs: Flow rate and Inlet Temperature Outputs: Parameter of Each Equipment 1: Procedure Structure of ANN Model 2: Unit = Min/Max number: 100~1000, step = 100 3: Input layer = neurons (Unit), number of inputs, activation function = ‘relu’, kernel initializer = ‘random uniform’ 4: Structure of Layer for Tuning for i in range (Min/Max number of layers: 1/5) 5: Min/Max number of Neurons: 25/ Unit , Neurons per layer: 25 → X End X_{25}^{25n} 6: Hidden layer = neurons(X_{25}^{25n}), activation function = ‘relu’, dropout rate , kernel initializer = ‘he uniform’ 7: Output layer = number of outputs, activation function = ‘relu’ 8: Model Compile: optimizer = ‘adam’, learning rate , loss function = ‘mean squared logarithmic error’ 9: Training: Hyper-parameters tuning using Bayesian optimization for j in range (trials: 5, execution per trial: 3) 10: dropout rate = [0.0, 0.1, 0.2, 0.3, 0.4, 0.5], learning rate = [1e-3, 1e-4, 1e-5] 11: Checking the condition for stopping the training if validation loss = min: save the ANN Model to ‘best model.h5’ else: j+1

An automatic hyper-parameter optimization algorithm was created such that the optimal number of hidden layers and neurons can be appropriately calculated for each sub-model through Bayesian optimization (lines 2–6 and 9–10 in Table 2). The number of hidden layers was at most 5, and the number of neurons in each layer was between 25 and 1,000. For the activation function, “relu” was applied

to all layers to minimize the shape transformation of the target value; in addition, the loss function was the “mean squared logarithmic error,” and “Adam” was used as the gradient descent technique. The number of hidden layers and neurons that minimize the errors for each sub-model were calculated. The hyper-parameter was searched seven times, and the model with the highest accuracy was selected as the final fitting model after two verifications. The reason for choosing such an automatic tuning method is to avoid a manual optimization of the model when data change and to find the hyper-parameter using quantitative judgment.

Modeling Results

The characteristics of the parameters were defined before verifying the modeling results of the entire system, the characteristics of the parameters were defined. Table 3 lists the minimum and maximum ranges of the calculated time-varying parameters. There were no negative numbers in the parameters, and positive values were calculated linearly using the activation function ‘relu’ in the MLP model. In addition, optimized MLP models were calculated for each parameter to create and fit the models with minimal validation losses and no overfitting.

Table 3: Range of estimated parameter.

Contents	2016		2017	
	Min	Max	Min	Max
Chiller01 $Pmt_{T_{out}} (kW \cdot ^\circ C)$	0.90	19.14	0.42	19.74
Chiller01 $Pmt_{COP} (-)$	0.75	1.13	0.23	0.97
Chiller02 $Pmt_{T_{out}} (kW \cdot ^\circ C)$	0.36	14.47	0.38	32.14
Chiller02 $Pmt_{COP} (-)$	0.26	2.33	0.69	1.12
Pump01 $Pmt_{Pump} (m)$	9.16	47.99	9.16	47.99
Pump02 $Pmt_{Pump} (m)$	9.60	47.99	9.59	47.99
Pump03 $Pmt_{Pump} (m)$	9.26	64.41	9.26	64.41
Pump04 $Pmt_{Pump} (m)$	9.16	34.91	9.16	34.91
Pump05 $Pmt_{Pump} (m)$	8.40	135.45	8.40	66.86
$Pmt_{HEX} (kcal)$	456.13	1322.52	496.83	1360.26

To check the accuracy of the device modeling method, we first conducted the calculations for each model. The parameter fitting results for a single model are shown in Table 4, and the results applied to the physical equations using the fitted parameters are listed in Table 5. In general, the value of R^2 was high for all parameters, although the RMSE value was relatively low. This is thought to be because, when the absolute value of the parameter is large, the absolute value of the error also increases; thus, a relatively small error is calculated as an extremely large number. In other words, in the case of a heat exchanger, because the parameter exceeds 900, an error of 30 between 900 and 930 results in a large RMSE value. Therefore, it is important to judge the accuracy by

substituting time-varying parameters in the physical equation. As shown in Table 5, the performance accuracy of each device is generally high. In particular, the power of the pump and the outlet temperature of the high and low temperatures of the heat exchanger have high accuracies in both the RMSE and R^2 . By contrast, in the case of the chiller, the accuracy of the COP parameter was low, which soon affected the power value, resulting in a low accuracy of the predicted power value in both chillers 01 and 02. This was confirmed to be due to the predicted data for the COP parameter in Table 2 having values outside the range of the training period. In the future, it will be necessary to consider how to increase the accuracy of the model by adding input variables with a significant influence (e.g., control signals) and increasing the amount of training data.

Result of System

The TRNSYS result was regarded as a value corresponding to the measured value, and the predicted result for each device and the result of calculation in the entire system are shown.

Table 4: Accuracy of Predicted Parameter.

Contents	RMSE	R^2
Chiller01 $Pmt_{T_{out}} (kW \cdot ^\circ C)$	0.432	0.972
Chiller01 $Pmt_{COP} (-)$	1.402	0.889
Chiller02 $Pmt_{T_{out}} (kW \cdot ^\circ C)$	0.587	0.999
Chiller02 $Pmt_{COP} (-)$	0.974	0.971
Pump01 $Pmt_{Pump} (m)$	1.119	0.994
Pump02 $Pmt_{Pump} (m)$	0.623	0.998
Pump03 $Pmt_{Pump} (m)$	1.351	0.994
Pump04 $Pmt_{Pump} (m)$	0.660	0.996
Pump05 $Pmt_{Pump} (m)$	4.434	0.894
$Pmt_{HEX} (kcal)$	83.661	0.985

Table 5: Accuracy of Performance Result

Contents	RMSE	R^2
Chiller01 $T_{out}^{out} (^\circ C)$	0.407	0.989
Chiller01 COP(-)	0.325	0.992
Chiller01 Power(kW)	3.729	0.914
Chiller02 $T_{out}^{out} (^\circ C)$	0.321	0.993
Chiller02 COP(-)	0.802	0.911
Chiller02 Power(kW)	3.515	0.994
Pump01 Power(kW)	0.007	0.992
Pump02 Power(kW)	0.013	0.999
Pump03 Power(kW)	0.005	0.998
Pump04 Power(kW)	0.004	0.998
Pump05 Power(kW)	0.022	0.966
HEX $T_{h2} (^\circ C)$	0.345	0.958
HEX $T_{c2} (^\circ C)$	0.374	0.982

As a result of single device modeling, the chilled water temperature of 6°C is generally discharged, but in the case of the system, the chilled water temperature is higher than that. The reason is that the flow rate does not fluctuate by temperature in the current system calculation process, and it is determined that the correct chilled water outlet temperature parameter is not fitted by substituting the measured value as it is. In the future, the flow rate sub-model will be added to improve the prediction accuracy. On the other hand, COP and power consumption have

high accuracy in both the device single model and the entire system prediction result.

Figure 7 shows the outlet temperature on the low temperature side of the heat exchanger, and the single device calculation result has much higher accuracy than the system calculation result. In the current heat storage tank model, gray box modeling that reflects heat loss and time-varying parameters is not implemented, so it is necessary to have high performance during system calculation along with modeling the heat storage tank in

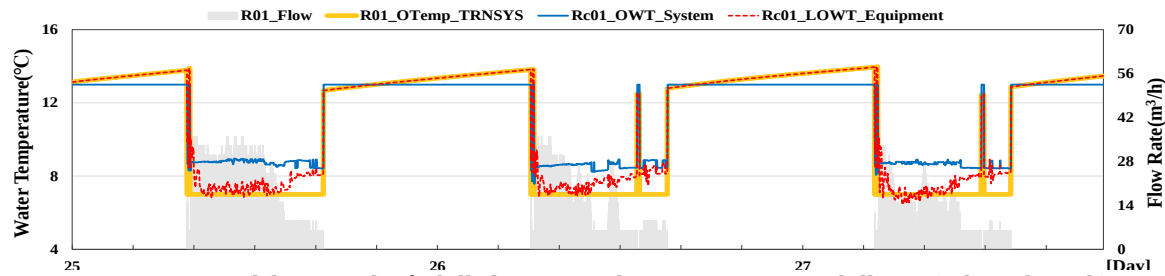


Figure 4 : Modeling Result of Chilled Water Outlet Temperature in Chiller01 (July 25th~27th)

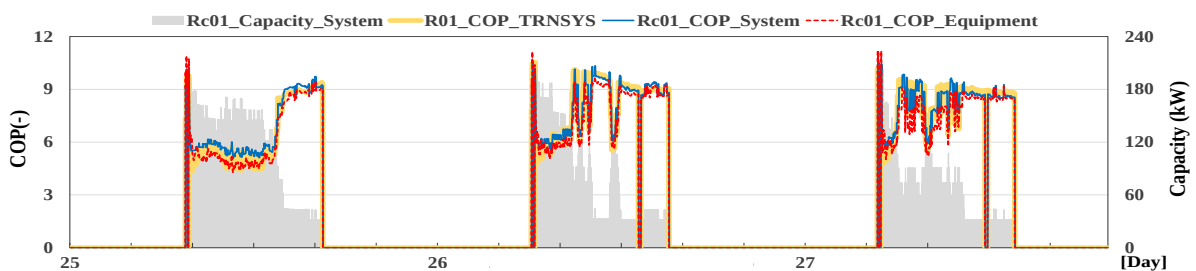


Figure 5 : Modeling Result of COP in Chiller01 (July 25th~27th)

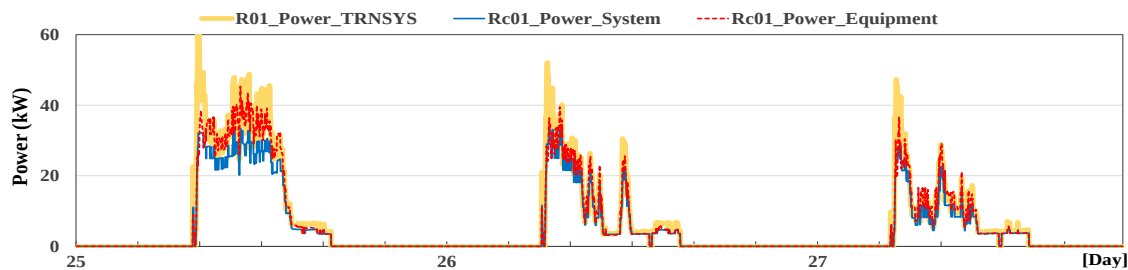


Figure 6 : Modeling Result of Power Consumption in Chiller01 (July 25th~27th)

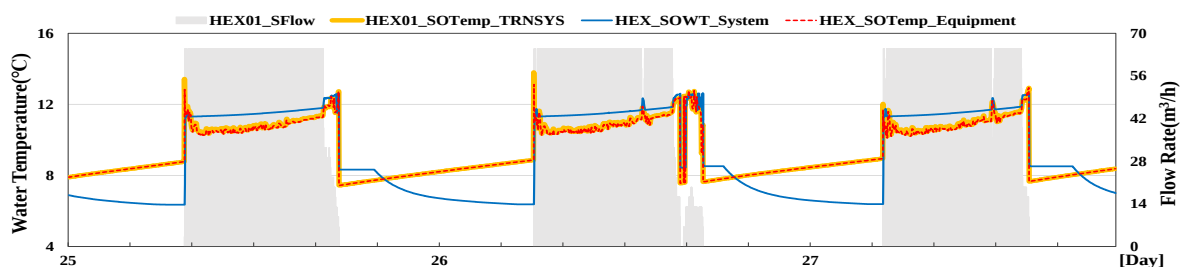


Figure 7 : Modeling Result of Low-Side Water Outlet Temperature in HEX (July 25th~27th)

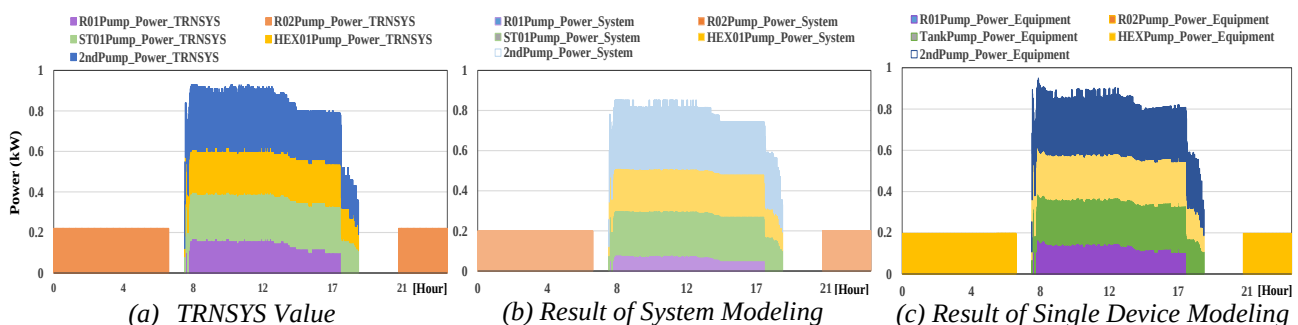


Figure 8: Modeling Result of Power Consumption in Pump 01~05 (July 25th)

the future. As a result of calculating the total power consumption of the heat source equipment and pump, the TRNSYS value for 3 months was about 6,572GWh, the system calculation result was about 7,495GWh, and the total sum of the power consumption in single device result was about 6,059GWh. As a result of the system calculation, the power consumption of Chiller 02 was about 20.3% higher than the measured value, indicating that the total power consumption was about 14% higher. This is largely due to the decrease in the COP prediction accuracy of Chiller 02, which performs heat charging operation. In the future, we intend to increase the accuracy of modeling by adding set temperature and operation control signals based on the results of a single device. Figure 8 compares the actual measurements, system calculation results, and single unit calculation results for 5 pumps on July 25th. For the five pumps, the total power consumption of the system and single unit differs by approximately 1.8% and -4.3 %. Therefore, the modeling method of pumps and heat exchangers is considered to be highly accurate.

Conclusion

A simulation tool was developed to automatically simulate HVAC&R systems in commercial buildings using data collected from real buildings. In addition, a gray-box modeling method was introduced to employ the methods, and a calculation method was developed that minimizes the number of parameters and allows the parameters to respond flexibly when outside conditions change. After individual device modeling, calculations were conducted for the entire system. As a result of the modeling, both the single-device modeling and system modeling results have a high accuracy. For system modeling, it is necessary to find a way to make the parameters more accurate, but we will be able to predict the performance of the device by fitting time-varying parameters. In the future, we will create control and flow rate models to monitor the operation of the system using the data and improve the performance of the sub-model to reduce the workload and develop a gray-box model with a relatively high accuracy.

References

- Sawant, P., Bürger, A., Doan, M. D., Felsmann, C., Pfafferoth, J. (2020). Development and experimental evaluation of grey-box models of a microscale polygeneration system for application in optimal control. *Energy and Buildings* 215, 1-12.
- Coninck, R. D., Magnusson, F., Åkesson, J., Helsen, L. (2014). Grey-Box Building Models for Model Order Reduction and Control. *Proceedings of the 10th International ModelicaConference*. Sweden(Lund), 10-12 March 2014.
- Takagi, Y. (2015) The Characteristic Analysis of a University Building: Dynamics Modelling by Grey Box Approach. *SICE Journal of Control* 8, 328–333.
- Braun, J. E., Chaturvedi, N. (2002) An Inverse Grey-box Model for Transient Building Load Prediction. *HVAC&R RESEARCH* 8, 73–99.
- Afram, A., Janabi-Sharifi, F. (2014) Review of modeling methods for HVAC systems. *Applied Thermal Engineering* 67, 507–519.
- Afram, A., Janabi-Sharifi, F. (2015) Grey-box modeling and validation of residential HVAC system for control system design. *Applied Energy* 137, 134–150.
- Lee, T., Lu, W. (2010) An evaluation of empirically-based models for predicting energy performance of vapor-compression water chillers. *Applied Energy* 87, 3486–3493.
- Hydeman, M., Webb, N., Sreedharan, P., Blanc, S. (2002) Development and Testing of a Reformulated Regression-Based Electric Chiller Model. *2002 Annual Conference*, Honolulu(HI), 1-10.
- Estrada-Flores, S., Merts, I., De Ketelaere, B., Lammertyn, J. (2006) Development and validation of “grey-box” models for refrigeration applications: a review of key concepts. *International Journal of Refrigeration* 29, 931–946.
- Berkenkamp, F., Gwerder, M. (2014) Hybrid model predictive control of stratified thermal storages inbuildings. *Energy and Buildings* 84, 233–240.
- Arahal, M. R., Cirre, C. M., Berenguel, M. (2008) Serial grey-box model of a stratified thermal tank for hierarchical control of a solar plant. *Solar Energy* 82, 441–451.
- Tardioli, G., Narayan, A., Kerrigan, R., Oates, M., O'Donnell, J., Finn, D.P. (2020) A methodology for calibration of building energy models at district scale using clustering and surrogate techniques. *Energy and Buildings* 226, 1-24.
- Singh, K., Das, R. (2016) A feedback model to predict parameters for controlling the performance of a mechanical draft cooling tower. *Applied Thermal Engineering* 105, 519–530.
- Pugliese, F., De Paola, F., Fontana, N., Giugni, M., Marini, G. (2016) Experimental characterization of two Pumps As Turbines for hydropower generation. *Renewable Energy* 99, 180–187.
- Harish, V.S.K.V., Kumar, A. (2016) A review on modeling and simulation of building energy systems. *Renewable and Sustainable Energy Reviews* 56, 1272–1292.
- Gordon, J.M., Ng, K. C. (1995) Predictive and diagnostic aspects of a universal thermodynamic model for chillers. *International Journal of Heat and Mass Transfer* 38, 807–818.
- ABB Group. Daikin Industries Ltd., Elster Group GmbH, GridPoint, Inc., Jones Lang LaSalle, Siemens AG (edited by). (2019). *Energy Management Systems Market Size, Share & Trends Analysis Report By Product, By Component, By Solution, By Vertical, By End Use, By Region, And Segment Predicts, 2019 - 2025*. Routledge. London (UK).
- TRNSYS 18 Volume 4 Mathematical Reference (2017) *The Board of Regents of the University of Wisconsin System*, 1